

QMS Journal Club

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QMS Journal Club 1st meeting: Linear algebra

A vector $\vec{v}$ is an n -tuple of entries, which can be represented by a column vector,

$$\vec{v} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}. \quad (1)$$

For our purposes, we focus only on complex vector space, which is the space of all complex vectors. The n -dimensional complex vector space is written as $\mathbb{C}^n$.

Vector space has two operations,

- 1 Vector addition, $\vec{v} = \vec{v}_1 + \vec{v}_2$
- 2 Scalar multiplication, $c\vec{v}$

A quantum bit (qubit) is a superposition (linear combination) of the basis states ($|0\rangle$, $|1\rangle$),

$$|\psi\rangle = \psi_0|0\rangle + \psi_1|1\rangle, \quad (2)$$

where ψ_0 and ψ_1 are called probability amplitudes, which are generally complex. The notation $|\psi\rangle$ is called a ket vector and can be represented as a column vector,

$$|\psi\rangle = \begin{pmatrix} \psi_0 \\ \psi_1 \end{pmatrix}, \quad (3)$$

where $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. $|0\rangle$ and $|1\rangle$ are orthonormal to each other.

The dual to the ket vector is a bra vector,

$$\langle\psi| = \bar{\psi}_0\langle 0| + \bar{\psi}_1\langle 1| = (\bar{\psi}_0 \quad \bar{\psi}_1), \quad (4)$$

where the overhead bar is complex conjugate operation.

The state vectors of a qubit live in the Hilbert space, which is a two-dimensional complex vector space $\mathcal{H} = \mathbb{C}^2$. The Hilbert space is equipped with an inner product operation,

$$\begin{aligned}\langle\psi|\psi\rangle &= (\bar{\psi}_0 \quad \bar{\psi}_1) \begin{pmatrix} \psi_0 \\ \psi_1 \end{pmatrix} \\ &= |\psi_0|^2 + |\psi_1|^2.\end{aligned}\tag{5}$$

There is also an outer product (tensor product),

$$\begin{aligned}|\psi\rangle\langle\psi| &= \begin{pmatrix} \psi_0 \\ \psi_1 \end{pmatrix} (\bar{\psi}_0 \quad \bar{\psi}_1) \\ &= \begin{pmatrix} |\psi_0|^2 & \psi_0\bar{\psi}_1 \\ \psi_1\bar{\psi}_0 & |\psi_1|^2 \end{pmatrix}\end{aligned}\tag{6}$$

There exists four extremely important matrices, called the Pauli matrices:

$$\sigma_0 = I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad (7)$$

$$\sigma_1 = X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad (8)$$

$$\sigma_2 = Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad (9)$$

$$\sigma_3 = Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (10)$$

It is important to take note that

$$XY = iZ; YX = -iZ, \quad (11)$$

$$YZ = iX; ZY = -iX, \quad (12)$$

$$ZX = iY; XZ = -iY, \quad (13)$$

$$XX = YY = ZZ = I. \quad (14)$$

The Pauli matrices are traceless, i.e. the sum of diagonal elements is zero, $\text{Tr}(\sigma_i) = 0$.

It is also Hermitian, i.e. its conjugate transpose is equivalent to itself, $\sigma_i^\dagger = \sigma_i$, where $\dagger$ denotes conjugate transpose operation.

Let

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}; |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad (15)$$

$$|+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}; |-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad (16)$$

$$|+i\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}; |-i\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}. \quad (17)$$

The Pauli matrices have two eigenvalues ± 1 ,

$$X|+\rangle = |+\rangle; X|-\rangle = -|-\rangle, \quad (18)$$

$$Y|+i\rangle = |+i\rangle; Y|-i\rangle = -|-i\rangle, \quad (19)$$

$$Z|0\rangle = |0\rangle; Z|1\rangle = -|1\rangle. \quad (20)$$

The Pauli matrices can be formulated by outer products,

$$X = |+\rangle\langle+| - |-\rangle\langle-|, \quad (21)$$

$$Y = |+i\rangle\langle+i| - |-i\rangle\langle-i|, \quad (22)$$

$$Z = |0\rangle\langle 0| - |1\rangle\langle 1|. \quad (23)$$

Previously, we mentioned that $\langle\psi|$ is the dual to $|\psi\rangle$. By using conjugate transpose operation, they are related by

$$\langle\psi| = (|\psi\rangle)^\dagger. \quad (24)$$

If A and B are two matrices, then $(AB)^\dagger = B^\dagger A^\dagger$. Therefore, $(A|\psi\rangle)^\dagger = \langle\psi|A^\dagger$.

Also, $(A + B)^\dagger = A^\dagger + B^\dagger$. Using this fact, one can easily show that the Pauli matrices are Hermitian, $\sigma_i^\dagger = \sigma_i$.

A matrix is unitary if $U^\dagger = U^{-1}$. The Pauli matrices are Hermitian and unitary.

Tensor product manifests itself in matrices as Kronecker product. Given two matrices A and B ,

$$\begin{aligned} A \otimes B &= \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \otimes \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} \\ &= \begin{pmatrix} a_{11} \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} & a_{12} \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} \\ a_{21} \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} & a_{22} \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} \end{pmatrix} \\ &= \begin{pmatrix} a_{11}b_{11} & a_{11}b_{12} & a_{12}b_{11} & a_{12}b_{12} \\ a_{11}b_{21} & a_{11}b_{22} & a_{12}b_{21} & a_{12}b_{22} \\ a_{21}b_{11} & a_{21}b_{12} & a_{22}b_{11} & a_{22}b_{12} \\ a_{21}b_{21} & a_{21}b_{22} & a_{22}b_{21} & a_{22}b_{22} \end{pmatrix}. \end{aligned} \quad (25)$$

In general, $A \otimes B \neq B \otimes A$.

If we have three vectors $|v_1\rangle, |v_2\rangle \in V$ and $|w\rangle \in W$,

$$(|v_1\rangle + |v_2\rangle) \otimes |w\rangle = |v_1\rangle \otimes |w\rangle + |v_2\rangle \otimes |w\rangle = |v_1 w\rangle + |v_2 w\rangle, \quad (26)$$

$$|w\rangle \otimes (|v_1\rangle + |v_2\rangle) = |w\rangle \otimes |v_1\rangle + |w\rangle \otimes |v_2\rangle = |wv_1\rangle + |wv_2\rangle. \quad (27)$$

If A acts on the vector space V and B acts on the vector space W , then

$$(A \otimes B)|v_1 w\rangle = A|v_1\rangle \otimes B|w\rangle. \quad (28)$$

Question: The Hadamard operator on one-qubit can be written as

$$H = \frac{1}{\sqrt{2}} [(|0\rangle + |1\rangle)\langle 0| + (|0\rangle - |1\rangle)\langle 1|]. \quad (29)$$

Show that the Hadamard transform on two qubits can be written as

$$H^{\otimes 2} = H \otimes H = \frac{1}{2} \sum_{x_1, x_2, y_1, y_2 \in \{0, 1\}} (-1)^{x_1 y_1 + x_2 y_2} |x_1 x_2\rangle \langle y_1 y_2|. \quad (30)$$

$$\begin{aligned}
H \otimes H &= \frac{1}{\sqrt{2}} [(|0\rangle + |1\rangle)\langle 0| + (|0\rangle - |1\rangle)\langle 1|] \otimes \\
&\quad \frac{1}{\sqrt{2}} [(|0\rangle + |1\rangle)\langle 0| + (|0\rangle - |1\rangle)\langle 1|] \\
&= \frac{1}{2} [(|00\rangle + |01\rangle + |10\rangle + |11\rangle)\langle 00| \\
&\quad + (|00\rangle - |01\rangle + |10\rangle - |11\rangle)\langle 01| \\
&\quad + (|00\rangle + |01\rangle - |10\rangle - |11\rangle)\langle 10| \\
&\quad + (|00\rangle - |01\rangle - |10\rangle + |11\rangle)\langle 11|] \\
&= \frac{1}{2} \sum_{x_1, x_2, y_1, y_2 \in \{0, 1\}} (-1)^{x_1 y_1 + x_2 y_2} |x_1 x_2\rangle \langle y_1 y_2|.
\end{aligned}$$

The trace of a matrix A is defined as the sum of its diagonal elements,

$$\begin{aligned} \text{Tr}(A) &= \text{Tr}(|A|) = \text{Tr} \left(\sum_{jk} |j\rangle \langle j| A |k\rangle \langle k| \right) \\ &= \text{Tr} \left(\sum_{jk} A_{jk} |j\rangle \langle k| \right) = \sum_{ijk} A_{jk} \langle i|j\rangle \langle k|i\rangle = \sum_{ijk} A_{jk} \langle k|i\rangle \langle i|j\rangle \\ &= \sum_{jk} A_{jk} \langle k|j\rangle = \sum_{jk} A_{jk} \delta_{kj} = \sum_j A_{jj}. \end{aligned} \quad (31)$$

The trace of a matrix satisfies a few properties:

- ① $\text{Tr}(AB) = \text{Tr}(BA)$
- ② $\text{Tr}(ABC) = \text{Tr}(BCA) = \text{Tr}(CAB)$
- ③ $\text{Tr}(A + B) = \text{Tr}(A) + \text{Tr}(B)$
- ④ $\text{Tr}(cA) = c\text{Tr}(A)$
- ⑤ $\text{Tr}(A^T) = \text{Tr}(A)$
- ⑥ $\text{Tr}(A^T B) = \text{Tr}(AB^T) = \text{Tr}(B^T A) = \text{Tr}(BA^T)$
- ⑦ $\text{Tr}(A \otimes B) = \text{Tr}(A)\text{Tr}(B)$

Suppose $|\psi\rangle$ is a unit vector and A is an arbitrary operator.

$$\begin{aligned} \text{Tr}(A|\psi\rangle\langle\psi|) &= \sum_i \langle i|A|\psi\rangle\langle\psi|i\rangle \\ &= \sum_i \langle\psi|i\rangle\langle i|A|\psi\rangle \\ &= \langle\psi|A|\psi\rangle. \end{aligned} \tag{32}$$

QMS Journal Club 2nd meeting: Postulates of quantum mechanics and Bloch sphere I

Postulates of quantum mechanics

Postulate 1: Associated to any isolated physical system is a complex vector space with inner product, i.e. a Hilbert space, known as the state space of the system. The system is completely described by its state vector, which is a unit vector in the systems' state space.

Postulate 2: The evolution of a closed quantum system is described by a unitary transformation. That is, the state $|\psi\rangle$ of the system at time t_1 is related to the state $|\psi'\rangle$ of the system at time t_2 by a unitary operator U which depends only on the times t_1 and t_2 ,

$$|\psi'\rangle = U|\psi\rangle. \quad (33)$$

Explicitly, it can be described by the Schrodinger equation,

$$i\hbar \frac{d}{dt} |\psi\rangle = \hat{H} |\psi\rangle. \quad (34)$$

Or,

$$|\psi(t_2)\rangle = e^{\frac{-i\hat{H}(t_2-t_1)}{\hbar}}|\psi(t_1)\rangle. \quad (35)$$

Question: Suppose A and B are commuting Hermitian operators. Show that $e^A e^B = e^{(A+B)}$ by expanding the left and right hand sides until $n = 3$.

Question: Show that e^{iK} is unitary for some Hermitian matrix K . You only need to expand until $n = 6$.

Note that the matrix exponential is given by

$$e^A = \sum_{n=0}^{\infty} \frac{A^n}{n!}. \quad (36)$$

Postulate 3: Quantum measurements are described by a collection $\{M_m\}$ of measurement operators. These are operators acting on the state space of the system being measured. The index m refers to the measurement outcomes that may occur in the experiment. If the state of the quantum system is $|\psi\rangle$ immediately before the measurement then the probability that result m occurs is given by

$$p(m) = \langle\psi|M_m^\dagger M_m|\psi\rangle, \quad (37)$$

and the state of the system after the measurement is

$$\frac{M_m|\psi\rangle}{\sqrt{\langle\psi|M_m^\dagger M_m|\psi\rangle}}. \quad (38)$$

The measurement operators satisfies the completeness equation,

$$\sum_m M_m^\dagger M_m = I. \quad (39)$$

A projective measurement is described by an observable, M , a Hermitian operator on the state space of the system being observed. The observable can be written as

$$M = \sum_m m |m\rangle \langle m|, \quad (40)$$

where $P_m = |m\rangle \langle m|$ is the projector onto the eigenspace of M with eigenvalue m , satisfying $P_m P_{m'}^\dagger = \delta_{mm'} P_m$.

The expected value of the measurement is

$$\begin{aligned} E(M) &= \langle M \rangle = \sum_m m p(m) \\ &= \sum_m m \langle \psi | P_m | \psi \rangle \\ &= \langle \psi | \sum_m m |m\rangle \langle m| | \psi \rangle \\ &= \langle \psi | M | \psi \rangle \end{aligned} \quad (41)$$

Question: The variance associated to observations of M , $[\Delta(M)]^2$ is given by

$$[\Delta(M)]^2 = \langle (M - \langle M \rangle)^2 \rangle. \quad (42)$$

Show that it is equivalent to $[\Delta(M)]^2 = \langle M^2 \rangle - \langle M \rangle^2$.

Question: Suppose A and B are two Hermitian operators, and $\langle \psi | AB | \psi \rangle = x + iy$. Show that $\langle \psi | [A, B] | \psi \rangle = 2iy$, $\langle \psi | \{A, B\} | \psi \rangle = 2x$.

The above implies that

$$|\langle \psi | [A, B] | \psi \rangle|^2 + |\langle \psi | \{A, B\} | \psi \rangle|^2 = 4|\langle \psi | AB | \psi \rangle|^2. \quad (43)$$

By the Cauchy-Schwarz inequality,

$$|\langle \psi | AB | \psi \rangle|^2 \leq \langle \psi | A^2 | \psi \rangle \langle \psi | B^2 | \psi \rangle. \quad (44)$$

Hence,

$$|\langle \psi | [A, B] | \psi \rangle|^2 \leq 4 \langle \psi | A^2 | \psi \rangle \langle \psi | B^2 | \psi \rangle. \quad (45)$$

Question: Suppose C and D are two observables, and $A = C - \langle C \rangle$, $B = D - \langle D \rangle$. Obtain the Heisenberg's uncertainty principle,

$$\Delta(C)\Delta(D) \geq \frac{|\langle \psi | [C, D] | \psi \rangle|}{2}. \quad (46)$$

QMS Journal Club 3rd meeting: Postulates of quantum mechanics and Bloch sphere II

Phase

Global phase factor is a term $e^{i\theta}$ attached to $|\psi\rangle$ without changing the measurement statistics of $|\psi\rangle$,

$$\langle\psi|e^{-i\theta}M_m^\dagger M_me^{i\theta}|\psi\rangle = \langle\psi|M_m^\dagger M_m|\psi\rangle. \quad (47)$$

For this reason, we say that $|\psi\rangle \sim e^{i\theta}|\psi\rangle$.

Relative phase is a term $e^{i\theta}$ attached between probability amplitudes,

$$|\psi\rangle = \alpha|\psi_\alpha\rangle + e^{i\theta}\beta|\psi_\beta\rangle. \quad (48)$$

Relative phase is a basis-dependent concept and may give rise to physically observable differences in measurement statistics.

For example, $|+\rangle = \frac{|0\rangle+|1\rangle}{\sqrt{2}}$ and $|-\rangle = \frac{|0\rangle-|1\rangle}{\sqrt{2}}$ are not the same up to a relative phase shift.

For one quantum bit (qubit), the Hilbert space is given as $\mathcal{H} = \mathbb{C}^2$, where

$$|\psi\rangle = \psi_0|0\rangle + \psi_1|1\rangle, \quad (49)$$

$$= \begin{pmatrix} \psi_0 \\ \psi_1 \end{pmatrix}, \quad (50)$$

where ψ_0 and ψ_1 are complex numbers, and $|\psi_0|^2 + |\psi_1|^2 = 1$.

At first glance, there are 4 degrees of freedom. However, due to the normalization condition and the equivalence of global phase factor, we only need 2 degrees of freedom to parametrize a one-qubit state,

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle. \quad (51)$$

Given the following Pauli matrices, find the expected values of Pauli- X , Y , and Z for $|\psi\rangle$.

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (52)$$

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right) |0\rangle + e^{i\phi} \sin\left(\frac{\theta}{2}\right) |1\rangle. \quad (53)$$

$$\langle\psi|X|\psi\rangle = \quad (54)$$

$$\langle\psi|Y|\psi\rangle = \quad (55)$$

$$\langle\psi|Z|\psi\rangle = \quad (56)$$

The expected values of Pauli matrices correspond to the spherical coordinates when the radius equals to 1,

$$\langle \psi | X | \psi \rangle = \sin \theta \cos \phi, \quad (57)$$

$$\langle \psi | Y | \psi \rangle = \sin \theta \sin \phi, \quad (58)$$

$$\langle \psi | Z | \psi \rangle = \cos \theta. \quad (59)$$

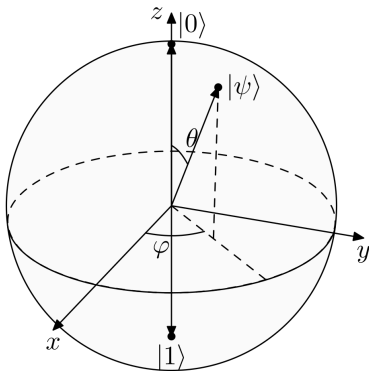


Figure 1: Bloch sphere

Postulates of quantum mechanics

Postulate 4: The state space of a composite physical system is the tensor product of the state spaces of the component physical systems. Moreover, if we have systems numbered 1 through n , and system number i is prepared in the state $|\psi_i\rangle$, then the joint state of the total system is

$$|\psi_1\rangle \otimes |\psi_2\rangle \otimes \dots \otimes |\psi_n\rangle. \quad (60)$$

Postulate 4 enables us to define quantum entanglement. Suppose we are given two qubit states, $|\psi\rangle \in \mathcal{H}_A = \mathbb{C}^2$ and $|\phi\rangle \in \mathcal{H}_B = \mathbb{C}^2$. The two-qubit state $|\psi_{AB}\rangle$ is entangled if $|\psi_{AB}\rangle \neq |\psi\rangle \otimes |\phi\rangle$.

For example, the following state $|\psi_{AB}\rangle$ is separable,

$$\begin{aligned} |\psi_{AB}\rangle &= \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle) \\ &= \frac{1}{2}[|0\rangle \otimes (|0\rangle + |1\rangle) + |1\rangle \otimes (|0\rangle + |1\rangle)] \\ &= \frac{1}{2}(|0\rangle + |1\rangle) \otimes (|0\rangle + |1\rangle) \\ &= |+\rangle \otimes |+\rangle \end{aligned}$$

It is not always obvious to identify if a given two-qubit state is separable or not. One can derive separability condition for two qubits. As long as the two-qubit state satisfies the separability condition, the two-qubit state is separable (else, it is entangled).

Let $|\psi\rangle = \psi_0|0\rangle + \psi_1|1\rangle$, $|\phi\rangle = \phi_0|0\rangle + \phi_1|1\rangle$. Then

$$\begin{aligned} |\psi_{AB}\rangle &= |\psi\rangle \otimes |\phi\rangle \\ &= (\psi_0|0\rangle + \psi_1|1\rangle) \otimes (\phi_0|0\rangle + \phi_1|1\rangle) \\ &= \psi_0\phi_0|00\rangle + \psi_0\phi_1|01\rangle + \psi_1\phi_0|10\rangle + \psi_1\phi_1|11\rangle \end{aligned}$$

Notice that the probability amplitudes of $|00\rangle$ and $|11\rangle$, when multiplied together, is the same as the product of probability amplitudes of $|10\rangle$ and $|01\rangle$. We can conclude that for a two-qubit state $|\psi_{AB}\rangle = \psi_{00}|00\rangle + \psi_{01}|01\rangle + \psi_{10}|10\rangle + \psi_{11}|11\rangle$, if $\psi_{00}\psi_{11} = \psi_{10}\psi_{01}$, the two-qubit state is separable. Else, it is entangled.

Can Alice communicate a number of classical bits of information by transmitting a smaller number of qubits, under the assumption that Alice and Bob receive an entangled resource beforehand?

- Alice and Bob initially share a pair of qubits in the Bell state,

$$|\psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}; \quad (61)$$

- If Alice wants to send 00, she does nothing; If Alice wants to send 01, she applies Z ; If Alice wants to send 10, she applies X ; If Alice wants to send 11, she applies iY ;

- The final two-qubit state becomes:

$$00 : |\psi\rangle \rightarrow \frac{|00\rangle + |11\rangle}{\sqrt{2}} \quad (62)$$

$$01 : |\psi\rangle \rightarrow \frac{|00\rangle - |11\rangle}{\sqrt{2}} \quad (63)$$

$$10 : |\psi\rangle \rightarrow \frac{|10\rangle + |01\rangle}{\sqrt{2}} \quad (64)$$

$$11 : |\psi\rangle \rightarrow \frac{|01\rangle - |10\rangle}{\sqrt{2}} \quad (65)$$

- Alice sends her qubit to Bob;
- Bob performs a Bell basis measurement to determine which of the four possible bit strings Alice sent.

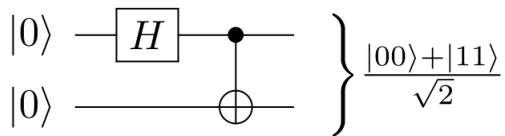


Figure 2: Bell state creation

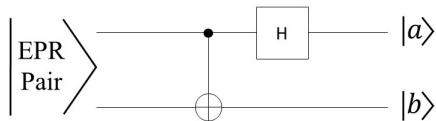


Figure 3: Bell basis measurement